

Sam Ethan Mathew

Cupertino, CA | semathew@andrew.cmu.edu | (669) 269-5144 | Github | LinkedIn

Education

Carnegie Mellon University - Pittsburgh, PA

May 2028

B.S. Electrical and Computer Engineering, Minor in Artificial Intelligence

- Relevant coursework: Discrete Math (21-127), Matrices & Linear Transformations (21-241), Intro ECE (18-100), Data Structures (15-122), Probability (21-325).
- Dual enrollment (De Anza College): Data Abstraction & Structures (CIS 22C), C++ Programming (CIS 22A/22B), Python Programming (CIS 41A).

Skills

Programming: C, C++, Python, Java, Go (exposure), JavaScript/TypeScript (Node, React, Vue), SQL

Systems/Infra: Linux/Unix, Git/GitHub, GitLab (exposure), Perforce (exposure), Docker, Kubernetes (learning), Jenkins (exposure), Ansible (learning), Microservices, Container/Cluster mgmt

GPU/AI Acceleration: ONNX Runtime, Real-time inference; CUDA (learning), cuDNN/cuBLAS (learning), NCCL (learning), TensorRT (learning)

ML/DL/CV: PyTorch/TensorFlow, scikit-learn, YOLO, LightFM; validation frameworks, data preprocessing, profiling & performance tuning

CS Topics: Data Structures & Algorithms, Distributed Systems, Operating Systems, Hardware Virtualization (concepts), Diagnostics, Tracing

Experience

Student Researcher, CMU Robotics Institute — Pittsburgh, PA

Aug 2025 – Present

- Implemented A* and heuristic variants; benchmarked vs. NetworkX to verify optimality and runtime baselines.
- Built reproducible evaluation harnesses in Python (OMPL) with tracing/diagnostics and performance tuning; produced cost-function plots across 50+ trials.
- Created path planning for polygonal environments; explored Polyanya-style search and cost functions.
- Accelerated RRT* via gate-guided sampling, reducing time-to-optimal by 10% (median across 15 trials).

Software Developer, ScottyLabs — Pittsburgh, PA

Aug 2025 – Present

- Real-time student activities platform with RSVP/moderation workflows; containerized services with metrics for tracing, diagnostics, and latency monitoring.
- Built LightFM (WARP) recommender fusing collaborative + profile features, boosting Recall@10 by 18% and NDCG@10 by 24% vs. popularity baseline.
- Implemented Google OAuth 2.0 and Google Calendar sync; automation scripts for deploys.

Software Engineer (Open Source Contributor), SoloTech — Mountain View, CA

Summer 2025

- Shipped speech → structured JSON → ROS2 actions pipeline; validated multi-command sequences in RViz with safety checks.
- Generalized motion control to SO-100 arm and TurtleBot; hardened URDF loading, command validation, and reproducible testing (validation framework; automation scripts).
- Seamlessly integrated into a CLI framework to integrate into over 10+ robotic systems currently used by over 200+ users.

Projects

BizGrow — Gamified AI Micro-Loan Incubator (HackCMU)

Sept 2025

- Designed quests → XP/coins → eligibility funnel with deterministic scoring and admin tooling.
- Next.js + Tailwind + shadcn/ui; Supabase Postgres/Auth/Storage; Dockerized services and CI hooks (Jenkins/GitHub Actions exposure).

ICU Stay-Level Synthetic Data Generator

2025

- Trained **SDV CTGAN/TVAE** with clinical constraints; realism checks (KS/χ^2) and privacy guards (NN distance, MIA).
- TSTR utility: AUROC within [2–5] pp of real-only; MIA AUC $\approx$ [0.50–0.60], re-id risk < [0.5] %.

Helmet Compliance Detector (YOLOv8 + OpenCV)

2024

- Curated dataset and trained; exported ONNX; $\sim$ 35 FPS real-time stream with timestamped detections using accelerated inference.
- Added profiling, tracing, and performance tuning; processed 36 hours/120 videos, detected 9.2k cyclist instances; **accuracy 91%**.

Awards & Activities

- First Place, JaneStreet x CMU Estimathon, out of 100+ students.
- First Place, CMU Scotty Labs Ideathon (Cursor Track), out of 50 teams
- Founding SW Mentor, RoboRoar FTC 30449 (2025–Present); FRC 2473 “Gold Strikers” Alumni/Electrical Lead
- FIRST Robotics “Excellence in Engineering” (2025)
- FBLA State: 2nd (Game & Simulation 2025, Web Dev 2024); Chapter President (250+ members, \$30k+ raised)